

RGPV B.TECH. FIRST YEAR REVISION

MODULE

Basic Electrical & Electronics Engineering (BT-104) • PYQ Comprehensive Analysis & Model Solutions (2020–2025)

1. Exam Pattern & Strategic Insights

Based on a multi-year longitudinal analysis of the Rajiv Gandhi Proudyogiki Vishwavidyalaya (RGPV) examination papers for **BT-104 (Grading System)**, the structural format remains strictly deterministic while internal question types show clear evolutionary trends:

- **Structure:** Every paper consists of exactly **8 comprehensive questions**, from which students must choose and attempt **any 5 questions**.
- **Marking Scheme:** The total marks are **70**. Each complete question carries **14 marks**, almost universally divided into two sub-parts: **(a) and (b)**, each evaluated at **7 marks**.
- **Recent Trend Shifts (2023–2025):** While earlier sessions (2020–2022) focused heavily on pure qualitative explanations and descriptive theory, the 2023, 2024, and 2025 papers reflect a structural shift toward **coupled analytical sequences**. A typical 14-mark block now pairs a fundamental derivation or theorem statement in part (a) with a complex numerical problem in part (b) that tests its exact practical implementation.
- **Strategic Focus Areas:** 100% of the papers contain complex multi-loop network numericals, single-phase AC parameters, transformer equivalent configurations, and induction motor slip diagnostics. Consistent practice on these specific areas guarantees high scoring performance.

UNIT NUMBER	UNIT TITLE	CORE HIGH-FREQUENCY FOCUS AREAS	APPROX. WEIGHTAGE
Unit 1	DC Network Analysis	Superposition Theorem, Thevenin's Theorem, Mesh/Nodal Equations	20% – 25% (1.5 to 2 Qs)
Unit 2	AC Circuits (1- Φ & 3- Φ)	Series RLC Resonance, Star-Delta line/phase relationships	20% (1.5 Qs)
Unit 3	Transformers & Inductance	OC/SC Test equivalent circuit parameters, Co-axial Solenoids (L, M)	20% (1.5 Qs)
Unit 4	Electrical Machines	3-Phase Induction Motor Torque-Slip Curve, DC Machine Construction	20% (1.5 Qs)
Unit 5	Basic Electronics & Logic	BJT α, β relationships, Full Adders, JK/SR Flip-Flops, De Morgan	20% (1.5 Qs)

UNIT 1: DC NETWORK ANALYSIS

A. Core Topic & Weightage Hierarchy

- **Superposition Theorem & Thevenin's Theorem:** Absolute High Priority. Present in 100% of analyzed historical sessions as full 7-mark components.
- **Nodal & Mesh Analysis:** High Priority. Embedded directly as specific procedural mandates for solving dual-source networks.
- **Star-Delta (Y-Δ) Transformations:** Medium Priority. Frequently featured as a targeted short note or embedded configuration.
- **Source Conversions & Ideal Sources:** Moderate Importance. Typically tested within qualitative sub-questions.

B. Categorized Question Bank (Original Wording)

1. Most Important / Highly Repeated Questions

- **MOST IMPORTANT** State the step-by-step procedural rules to solve a linear network using **Thevenin's Theorem**.
DEC 2025, JUN 2024, MODEL PAPER
Repetition Count: **3 Times**
- **MOST IMPORTANT** Find the current flowing through the **3 Ω** resistor using the **Superposition Theorem** in the given multi-source circuit configuration. [Circuit: 20V source, 5Ω resistor in loop 1; 40V source, 6Ω resistor in loop 2; sharing a central 3Ω branch]. DEC 2025
Repetition Count: **Highly Critical Pattern**
- **MOST IMPORTANT** State and precisely explain the **Superposition Theorem** with a suitable electrical circuit example. JUN 2025, JUN 2024, JUN 2020
Repetition Count: **3 Times**

2. Frequently Asked Questions

- **FREQUENTLY ASKED** Determine the equivalent resistance across the circuit terminals A and B for the given complex resistor network. [Bridge/ladder network configuration]. DEC 2025
Repetition Count: **2 Times**
- **FREQUENTLY ASKED** Using **Nodal Analysis**, calculate the branch current traversing through the **10 Ω** resistor in the network diagram below. [Circuit: Two active loops containing multiple voltage sources]. DEC 2024
Repetition Count: **Continuous Pattern**
- **FREQUENTLY ASKED** Apply **Mesh Current Analysis** to determine the current moving through the **5 Ω** resistor in the given multi-loop active configuration. JUN 2025, DEC 2023
Repetition Count: **2 Times**

3. Moderate to Rare Importance Questions

- **MODERATE** Differentiate between dependent and independent energy sources. Fully explain with neat schematic sketches how a practical voltage source is transformed into an equivalent current source.
DEC 2024, DEC 2023, JUN 2023

- **RARE BUT POSSIBLE** Discuss ideal voltage sources and ideal current sources using exact graphic representations. **JUN 2024**

Critical Analytical Formula Summary for Unit 1:

Thevenin Load Current: $I_L = V_{th} / (R_{th} + R_L)$

Delta to Star transformation: $R_1 = (R_{12} * R_{31}) / (R_{12} + R_{23} + R_{31})$

Star to Delta transformation: $R_{12} = R_1 + R_2 + (R_1 * R_2 / R_3)$

Common Numerical Blueprint: Always expect a 3-loop or 2-node circuit diagram combining an independent current source (e.g., 3 A or 4 A) and an independent voltage source (e.g., 15V, 20V, or 30V). You will be ordered to find a single branch current using either Superposition or Mesh analysis.

C. Most Expected Questions for the Upcoming Exam

1. State **Thevenin's Theorem**. Outline the systematic procedure to calculate the open-circuit voltage (V_{th}) and internal resistance (R_{th}) across an arbitrary pair of load terminals.
2. A linear network contains two active loops. Loop 1 has a 20V source with a $5\ \Omega$ series resistance, and Loop 2 has a 40V source with a $6\ \Omega$ series resistance. They share a common middle branch containing a $3\ \Omega$ resistor. Calculate the current in the middle branch using the **Superposition Theorem**.

UNIT 2: ALTERNATING CURRENT (AC) CIRCUITS

A. Core Topic & Weightage Hierarchy

- **Series RLC Circuit Dynamics:** Paramount High Priority. Demands calculation of total circuit impedance, active current, operational power factor, and drawing the voltage phasor diagram.
- **Three-Phase Balanced Systems:** High Priority. Derivations and line/phase validation for Star (Y) and Delta (Δ) balanced configurations.
- **Sinusoidal Waveform Parameters:** Medium Priority. Formal definitions and calculations for RMS value, Average value, Form Factor, and Peak Factor.

B. Categorized Question Bank

1. Most Important / Highly Repeated Questions

- **MOST IMPORTANT** A series RLC circuit containing a resistance of **12 Ω** , an inductance of **0.15 H**, and a capacitance of **100 μF** is connected across a **100V, 50Hz** supply. Calculate the total circuit impedance, circuit current, operating power factor, and draw the complete voltage phasor diagram. **DEC 2025**

Repetition Count: 4 Times (with minor value changes)

- **MOST IMPORTANT** Draw the phasor diagram of a 3-phase delta-connected balanced load and derive the mathematical relation between phase and line quantities for both voltages and currents.

JUN 2024, NOV 2022, JUN 2022

Repetition Count: 3 Times

2. Frequently Asked Questions

- **FREQUENTLY ASKED** Define and derive expressions for the following parameters of an alternating sinusoidal waveform: (i) R.M.S. Value, (ii) Average Value, (iii) Form Factor, and (iv) Peak Factor.

JUN 2025, DEC 2024, JUN 2024, DEC 2023, JUN 2023

Repetition Count: 5 Times

- **FREQUENTLY ASKED** A three-phase balanced load consists of three equal impedances connected in delta across a 3-phase **400V, 50Hz** supply. The line current drawn is **20A** at a lagging power factor of **0.3**. Determine the value of the active impedance connected in each phase loop. **DEC 2025, DEC 2024**

Repetition Count: 2 Times

Key Formulas for AC Circuits:

Inductive Reactance: $X_L = 2 * \pi * f * L$ | Capacitive Reactance: $X_C = 1 / (2 * \pi * f * C)$

Total Impedance: $Z = \sqrt{R^2 + (X_L - X_C)^2}$

Power Factor: $\cos \phi = R / Z$ | Circuit Current: $I = V / Z$

Balanced Star: $V_{Line} = \sqrt{3} * V_{Phase}$, $I_{Line} = I_{Phase}$

Balanced Delta: $V_{Line} = V_{Phase}$, $I_{Line} = \sqrt{3} * I_{Phase}$

C. Most Expected Questions for the Upcoming Exam

1. A series RLC configuration has $R = 10\ \Omega$, $L = 0.1\ H$, and $C = 150\ \mu F$ supplied by a $200V, 50Hz$ alternating line. Calculate the total circuit inductive reactance, capacitive reactance, complex impedance, operational current, and power factor. Draw its voltage phasor diagram.
2. Explain the concept of Phase Sequence. Derive the quantitative expressions showing that line voltage is $\sqrt{3}$ times the phase voltage in a balanced star-connected network.

UNIT 3: TRANSFORMERS & MAGNETIC CIRCUITS

A. Core Topic & Weightage Hierarchy

- **Transformer OC/SC Testing:** Highest Priority. Standardized numeric analysis to calculate the primary-referred equivalent circuit parameters (R_0, X_0, R_{e1}, X_{e1}).
- **Solenoid Inductance Derivations:** High Priority. Deriving mathematical models for self-inductance and mutual inductance of closely wound coaxial solenoid systems.
- **Ferromagnetic Material Metrics:** Medium Priority. Descriptive analysis of B-H loops, normal magnetization patterns, and hysteresis losses.

B. Categorized Question Bank

1. Most Important / Highly Repeated Questions

- **MOST IMPORTANT** Open-circuit and short-circuit tests performed on a single-phase transformer yielded the following experimental data:
 - **O.C. Test (Primary Side):** $V_0 = 200V, I_0 = 0.7A, W_0 = 20 W$
 - **S.C. Test (Secondary Side):** $V_s = 10V, I_s = 10A, W_s = 40 W$

Calculate the complete set of equivalent circuit parameters referred to the primary driving side.

DEC 2024, DEC 2023, JUN 2023

Repetition Count: 3 Times (Identical Numerical Values)

- **MOST IMPORTANT** Derive the exact mathematical expressions for the coefficient of self-inductance of a single isolated long solenoid coil, and the mutual inductance for a pair of closely wound co-axial solenoid cylinders.

DEC 2025, DEC 2024, JUN 2024, JUN 2023

Repetition Count: 4 Times

2. Frequently Asked Questions

- **FREQUENTLY ASKED** Discuss and explain the normal magnetization characteristics (B-H curve) of ferromagnetic materials and outline the physical significance of hysteresis loss loop geometry.

DEC 2025, DEC 2024, JUN 2024, DEC 2023, JUN 2020

Repetition Count: 5 Times

- **FREQUENTLY ASKED** Enumerate the various internal losses occurring in a power transformer. Show how these individual loss components can be minimized by structural modifications.

NOV 2022, JUN 2022

Repetition Count: 2 Times

Inductance & Transformer Testing Mathematical Framework:

Self Inductance: $L = (\mu_0 * \mu_r * N^2 * A) / l$

Mutual Inductance: $M = K * \sqrt{L_1 * L_2} = (\mu_0 * N_1 * N_2 * A) / l$

From O.C. Test: $R_0 = V_0 / I_c$ where $I_c = I_0 * \cos \phi_0$; $X_0 = V_0 / I_m$
where $I_m = I_0 * \sin \phi_0$

From S.C. Test: $R_{e2} = W_s / (I_s)^2$ | $Z_{e2} = V_s / I_s$ | $X_{e2} = \sqrt{(Z_{e2}^2 - R_{e2}^2)}$

C. Most Expected Questions for the Upcoming Exam

1. Derive the mathematical relationship linking self-inductance coefficients (L_1, L_2) and the mutual coupling inductance parameter (M). Define the term 'coupling coefficient'.
2. A **25 kVA, 2000/200V** power transformer has core (iron) losses of **350W** and full-load copper losses of **400W**. Calculate the operational efficiency of the transformer when working at full load and unity power factor.

UNIT 4: ELECTRICAL MACHINES

A. Core Topic & Weightage Hierarchy

- **3-Phase Induction Motor Operation:** Absolute High Priority. Requires explanation of the Rotating Magnetic Field (RMF) concept, drawing the complete Torque-Slip curve, and explaining why synchronous operation is impossible.
- **Induction Motor Rotor Dynamics:** High Priority. Numerical evaluation of synchronous speed, operating fractional slip, slip rpm, and rotor frequency.
- **DC Machine Construction & Control:** Medium Priority. Structural layout, armature parts, commutator functionality, and speed control paradigms.

B. Categorized Question Bank

1. Most Important / Highly Repeated Questions

- **MOST IMPORTANT** Draw and explain the complete **Torque-Slip characteristics curve** of a 3-phase induction motor. Clearly mark and demarcate the stable region, unstable region, and the maximum pull-out torque boundary. **DEC 2024, DEC 2023, JUN 2023, JUN 2022**
Repetition Count: 4 Times
- **MOST IMPORTANT** Explain the fundamental operational principle of a 3-phase induction motor. Provide a clear engineering justification explaining why this machine can never achieve true synchronous speed. **DEC 2024, JUN 2025, JUN 2022, MODEL PAPER**
Repetition Count: 4 Times

2. Frequently Asked Questions

- **FREQUENTLY ASKED** A three-phase, **50Hz**, 2-pole induction motor operates with a full-load shaft speed of **2860 rpm**. Calculate: (i) Synchronous speed, (ii) Operational fractional slip, and (iii) Induced rotor EMF frequency. **DEC 2025**
Repetition Count: Continuous Problem Style
- **FREQUENTLY ASKED** Provide a detailed circuit diagram to explain the physical construction, key component functions (field poles, armature, commutator), and working principles of a standard DC Motor. **DEC 2025, JUN 2025, JUN 2024, JUN 2020**
Repetition Count: 4 Times

Induction Motor Operational Formulas:

Synchronous Speed: $N_s = (120 * f) / P$ (in rpm)
Fractional Slip: $s = (N_s - N_r) / N_s$ | Rotor Speed: $N_r = N_s * (1 - s)$
Rotor Current Frequency: $f_r = s * f$ | Slip Speed = $N_s - N_r$

C. Most Expected Questions for the Upcoming Exam

1. A 3-phase, 6-pole, **50Hz** induction motor exhibits a fractional slip of **1%** at zero-load condition and increases to **3%** under full mechanical load. Compute: (i) Synchronous speed, (ii) Actual rotor speed at no load, (iii) Rotor speed at full load, and (iv) Frequency of induced rotor currents at full load.
2. What structural and operational differences distinguish a separately excited DC generator from a self-excited DC generator? Explain briefly.

UNIT 5: BASIC ELECTRONICS & DIGITAL LOGIC

A. Core Topic & Weightage Hierarchy

- **Bipolar Junction Transistors (BJT):** Highest Priority. Demands mathematical derivation of current gain parameters (α and β) and solving current distribution configurations.
- **Combinational Logic Circuits:** High Priority. Implementation, truth tables, and gate layouts for Half Adder and Full Adder circuits.
- **Sequential Circuits & Theorems:** Medium Priority. SR and JK Flip-Flop designs using NAND gates, and verification of De Morgan's Laws.

B. Categorized Question Bank

1. Most Important / Highly Repeated Questions

- **MOST IMPORTANT** Define the current gain terms α and β with respect to Bipolar Junction Transistors (BJT) and derive the algebraic relationship between them. For a transistor operating circuit where $\alpha = 0.98$, $I_{CBO} = 5 \mu A$, and base drive current $I_B = 100 \mu A$, calculate the exact collector current (I_C) and emitter current (I_E).
DEC 2025, DEC 2024, MODEL PAPER
Repetition Count: **3 Times (Identical Numerical Matrix)**
- **MOST IMPORTANT** Design a **Full Adder combinational circuit**. Construct its complete operational truth table, establish the Boolean logic expressions for Sum and Carry outputs, and draw its logic diagram using basic logic gates. JUN 2024, DEC 2023, JUN 2022, MODEL PAPER
Repetition Count: **4 Times**

2. Frequently Asked Questions

- **FREQUENTLY ASKED** State and rigorously prove **De Morgan's Theorems** for two independent digital input variables using truth table validation. DEC 2025, DEC 2023, JUN 2023, NOV 2022, JUN 2020
Repetition Count: **5 Times**
- **FREQUENTLY ASKED** Draw the logic schematic of an **S-R Flip-Flop** using cross-coupled NAND gates. Explain its underlying truth table, detailing the invalid/forbidden operating state. DEC 2025, JUN 2025, NOV 2022, JUN 2022
Repetition Count: **4 Times**
- **FREQUENTLY ASKED** Draw and explain the forward and reverse bias V-I operational characteristic curve of a standard silicon PN Junction Diode. JUN 2025, JUN 2023, JUN 2020
Repetition Count: **3 Times**

Electronic & Logic Gate Fundamental Relations:

BJT Current Balance: $I_E = I_B + I_C$

Collector Current with Leakage: $I_C = \alpha * I_E + I_{CBO}$ | $\beta = \alpha / (1 - \alpha)$

Full Adder Boolean Logic: $\text{Sum} = A \oplus B \oplus C_{in}$ | $\text{Carry}_{out} = (A \cdot B) + (B \cdot C_{in}) + (C_{in} \cdot A)$

De Morgan's Laws: $[A + B]' = A' \cdot B'$ and $[A \cdot B]' = A' + B'$

C. Most Expected Questions for the Upcoming Exam

1. Explain the physical operation and characteristic curves of a BJT configured in the Common Emitter (CE) layout. Detail the active, saturation, and cut-off operational zones.
2. Draw the logic diagram and explain the operational mechanics of a master-slave **J-K Flip-Flop circuit**. Show how it resolves the race-around limitation found in simpler configurations.

6. Master Review & Last-Minute Exam Blueprint

A. Definitive High-Frequency Topic Checklist

CORE CONCEPT CATEGORY	SPECIFIC ENGINEERING DELIVERABLES EXPECTED	HISTORICAL PROBABILITY
Linear Network Theorems	Solving 2-source or 3-source active mesh problems using Superposition and calculating Thevenin models.	100% (Every Paper)
RLC Series Diagnostics	Computing inductive reactance, capacitive reactance, complex impedance, and drawing the voltage phasor diagram.	95% (Highly Constant)
Transformer Equivalence	Calculating core parameters (R_0, X_0) from open-circuit data and series parameters from short-circuit tests.	90%
3- Φ Induction Motors	Drawing the Torque-Slip curve showing breakdown torque, stable region, and solving for slip and rotor frequency.	95%
BJT Multi-Gain Algebra	Deriving $\beta = \alpha/(1-\alpha)$ and solving current networks given base current and reverse leakage thresholds.	85%
Combinational Adders	Drawing Full Adders using logic gates and creating the complete binary truth table.	80%

B. High-Scoring Execution Strategies

- Phasor Diagrams:** When solving AC network or transformer questions, always draw the associated phasor diagrams. A properly drawn phasor diagram with correct phase orientation (leading/lagging angles) earns full marks even if there is a minor numerical error in subsequent steps.
- Theorem Explanations:** For Thevenin's or Superposition theorems, always accompany the word statement with a generic 2-loop schematic layout showing how independent sources are deactivated (short-circuiting voltage networks, open-circuiting current paths).
- Torque-Slip Curves:** Ensure your Unit 4 curves clearly show the three distinct operational modes: motoring zone ($0 \leq s \leq 1$), generating zone ($s < 0$), and braking zone ($s > 1$).

C. Last-Minute Revision Formulas & Shortcuts

- Resonance condition in RLC networks:** Occurs when $X_L = X_C$, meaning the total impedance simplifies to $Z = R$, making the power factor perfectly unitary ($\cos \Phi = 1$). Resonance frequency is given by $f_r = 1 / (2 \pi \sqrt{LC})$.
- Transformer Transformation Ratio (K):** Defined as $K = V_2 / V_1 = N_2 / N_1 = I_1 / I_2$. When transferring resistance from secondary to primary side, divide by K^2 : $R_{e1} = R_1 + R_2 / K^2$.
- BJT Current Gain Conversion Shortcut:** If $\alpha = 0.98$, then $\beta = 0.98 / (1 - 0.98) = 0.98 / 0.02 = 49$. If $\alpha = 0.99$, then $\beta = 99$. This simplifies complex collector calculations.